

Library Management System

1. Noun-Verb Analysis

Raw candidates:

Student, Library, Book, Student ID, Name, Username, Password, PDF, Login, Register, Open, Issue, Return, Edit, Logout.

Surviving classes:

- Student
- Library
- Book

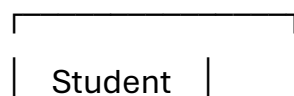
Discarded candidates:

- **Attributes filter:** Student ID, Name, Username, Password, Book Title, Author
- **UI/Implementation filter:** Login, Dashboard, PDF, Register
- **Operation/Verb filter:** Open, Issue, Return, Edit, Logout
- **Irrelevance/Redundancy filter:** System, Pending Book

2. CRC Cards

Class	Responsibilities	Collaborators
Student	Register, Login, View books, Issue/Return, Edit profile	Library, Book
Library	Manage books, Issue/Return books	Student, Book
Book	Store book details, Open PDF, Maintain availability	Library, Student

3. Domain Class Diagram



|-----|

| -studentId |

| -name |

| -username |

| -password |

|-----|

| +login() |

| +register() |

| +issueBook() |

| +returnBook() |

|-----|

| 1

|

| uses

|

| 1

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| Library |

|-----|

| -libraryId |

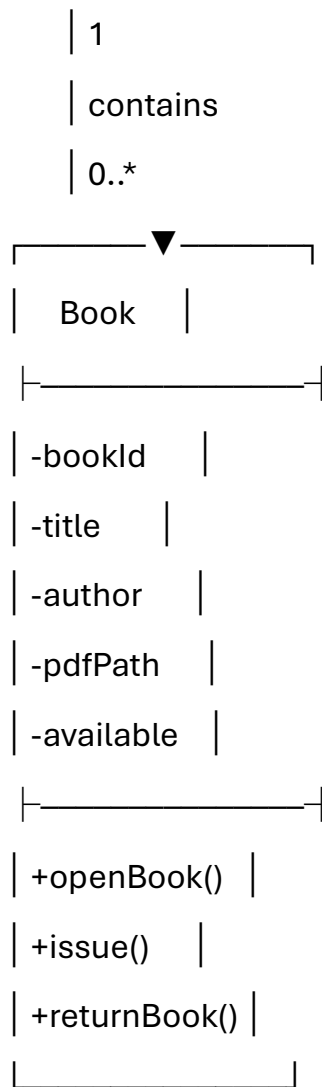
|-----|

| +addBook() |

| +issueBook() |

| +returnBook() |

|-----|



4. Object Diagram

student1 : Student

studentId = "IT001"

name = "Rahul"

username = "rahul01"

|

| uses

↓

library1 : Library

libraryId = "L001"

|

| contains

↓

book1 : Book

bookId = "B001"

title = "Design and Analysis of Algorithms"

pdfPath = "Algorithms.pdf"

available = false